

Central place theory and its reemergence in regional science

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Abstract This paper presents an interdisciplinary review—from the perspectives of geography, economics, and urban planning—of the literature on central place theory (CPT) and its important role in regional science. Along the way, CPT is compared to a more recent description/explanation of the space economy, the new economic geography (NEG). The main arguments of the paper are that for a host of reasons: (i) CPT is ripe for reemergence and; (ii) NEG models are best viewed as complements to, not competitors of, CPT. Some general observations for future research in regional science follow from these conclusions.

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1 Introduction

At any given time, the research of a discipline is both directed and motivated by a few path-breaking studies that serve as research paradigms. In the case of regional science, much recent thinking has followed ideas established by [Krugman \(1991a,b\)](#), who interconnected imperfect competition, externalities, and interregional trade to form a theoretical framework now labeled the new economic geography (NEG). This literature has provided an endogenous theory of regional growth—one where the economic landscape exhibits geographic specialization. For all of its success, the NEG has also attracted a fair amount of criticism—the concerns expressed by [Martin and Sunley \(1996\)](#) are especially well known—but most of this criticism fails to note that, like many of its rivals, the NEG is highly preoccupied with industrial production. As a result, the entire topic of consumer demand is essentially neglected, whether interest is in the access of households to private or to public goods. That deficiency, combined with the inherent mathematical complexities of the models, means that the economic landscapes of NEG tend to be highly simplistic. In any event, the equilibrated results typically lack the complex, multilevel attributes of real-world spatial-economic systems. As a remedy, this paper suggests that a reexamination of the merits of central place theory (CPT) is now in order. This corpus of thought, dismissed almost out of hand by the NEG pioneers (see, for example, comments by [Fujita et al. 1999](#), p. 27), accurately mirrors many features of real-world economic landscapes. The theory not only integrates, at least conceptually, a wide array of human activities but it also indicates how human processes are embedded in space or mediated by location. Moreover, CPT has very obvious practical implications for policy and, especially, urban and regional planning.

In essence, CPT consists of a set of assumptions and propositions that explain why hierarchically tiered centers are found at certain preferred locations on the economic landscape. Retail trade and service activities often tend to cluster, forming urban centers at one geographic scale and shopping centers at another. At both scales, activity clusters are ranked according to their relative centrality and functional complexity, and their frequency distribution is determined by the geographic scope of their drawing power. In this manner, a key feature of CPT is that basic patterns are consistently repeated across and within regions. Although not entirely developed, the framework informs many aspects of regional science—to name just a few: consumer shopping behavior; business location and clustering; economic base and input–output analysis; spatial diffusion and human movement; urban growth, rural–urban interdependence, regional development; and city-size distribution. CPT also holds concrete implications for urban and regional policy, including public facility location and for planning in cities and regions.

This paper provides an overview of CPT theory and its extensions and argues that the framework is experiencing—and deserves to experience—a reemergence as part of the core epistemology of regional science. It also very briefly distinguishes CPT from NEG models, but a complete review of NEG models is well beyond the present scope. While CPT has a number of key advantages, interest in it waned for about a 20-year span beginning in the early 1990s, when excitement among researchers shifted to NEG models. CPT was faulted for lacking microeconomic foundations, not

providing a general equilibrium approach, and in not being fully dynamic. Recently, however, a renewed interest in CPT has emerged due to its relative accessibility, plus its realism and value in predicting and describing empirical regularities such as Zipf's Law (Saichev et al. 2010), as well as the enduring stability of land use patterns and urban systems. Perhaps most important of all, CPT is invaluable for helping inform urban and regional economic development and planning strategies. Overall, this paper illustrates that CPT should be viewed as complementary to NEG and other urban system frameworks for describing how economic activity is organized across geographic space.

Toward that end, the discussion is organized as follows. To begin, Sects. 2 and 3 deliver a non-technical overview of traditional CPT and set out its three main concepts: (i) consumer choice; (ii) agglomeration; and (iii) a functional urban hierarchy. Next, Sect. 4 discusses how a hierarchy actually emerges within CPT and compares the framework with NEG. Section 5 describes the dynamic features of CPT, such as they are, and some recent empirical tests of its predictions, and Sect. 6 outlines some policy applications with an eye toward urban and regional planning. Section 7 presents conclusions drawn from the discussion and suggests some future research priorities.

2 The traditional theory

Comprehensive summaries of the ideas of Christaller (1933) and Lösch (1941), the two founders of CPT—along with various refinements and empirical extensions—include von Böventer (1963), Haggett (1965), Berry (1967), Beckmann (1968), Berry and Horton (1970), Parr and Denike (1970), Alao et al. (1977), Beavon (1977), Lloyd and Dicken (1977), King (1984), Mulligan (1984), and Berry and Parr (1988). What follows is a brief, nontechnical overview of the framework put forward by each.

2.1 The Christaller framework

Christaller (1933) used the concepts of the (population) threshold and the (geographic) range to provide a top-down explanation for the emergence of a cascading system of central places. Both of these concepts can be interpreted spatially—where the former is the inner range and the latter is the outer range—two attributes that serve as geographic signatures for each good provided in the system. The inner range encompasses sufficient customers for the typical business to break even (zero profits) and the outer range encompasses all those customers that travel to purchase the good provided. So any gap between the inner and outer ranges will generate a surplus or monopoly profit for the business. A continuum of goods and services is envisaged stretching from high-order items having large thresholds (specialized shopping goods) and ranges to low-order items having small thresholds and ranges (standardized convenience goods). The most central location in the entire system provides all of its goods and services—thereby satisfying the so-called exhaustive principle. But, moving down through this functional continuum, other locations on the landscape are sufficiently well located to provide some, but not all, goods that are provided at the most central location. These locations, which themselves differentially attract customers from nearby areas, are

termed central places. All of the central places are classified into a small number of orders, reflecting the system's hierarchical levels, which, in turn, reflect the number of bundled goods and services that each place provides.

The key feature of Christaller's system is the emergence of a regular hierarchy, wherein a central place on one level provides a bundle of goods and services that is specific to that level, as well as all bundles that are specific to all lower levels. As a result, central places on any given level serve a market area containing a number of lower-level places and rural areas. These (usually hexagonal) market areas are exactly nested within one another according to a factor K , which Christaller believed to be fixed in magnitude for a particular system. This spatial-functional arrangement, or multi-good equilibrium, was believed to be optimally efficient for serving a dispersed population—an implication that Hsu's recent (forthcoming) extension of the model demonstrates. The strengths of Christaller's scheme are its geometric simplicity and its insightful behavioral propositions, especially regarding consumer travel behavior. Figure 1 shows a typical CPT hierarchy, consisting of a spatially cascading set of market centers surrounded by nested hexagonal market areas.

The problem with Christaller's framework is that it is not convincing in terms of explaining the emergence of a realistic multi-good equilibrium. Basically, he claimed that spatial competition induces geometric packing as new firms erode the profits of earlier entrants. Presumably, location adjustments and firm entries and exits continue until the final spatial equilibrium is reached. In any case, this packing process tends to maximize the number of central places per unit of area, thereby minimizing overall customer transportation costs in the entire system.

2.2 The Lösch framework

Lösch, on the other hand, conceived a much more complex bottom-up central place system whose conceptual foundation is a triangular lattice of discrete farmsteads. He argued that Christaller's rigid hierarchy was too restrictive and, instead, proposed using hexagonal nets of market areas along a near-continuum of sizes, all of which focus on the system's most central location—the so-called metropolis. In essence, Lösch claimed that each of the goods in Christaller's functional continuum should have its own signature market area and that, as before, businesses or production points naturally arise at the center of each different-sized market area. But, in the Lösch framework, these nets are purposefully rotated in such a way that “city-rich” and “city-poor” sectors emerge, reflecting the varying numbers of production points that turn out to be coincident on the underlying lattice.

Lösch's thinking was that this transformation would create a system that would be most efficient both for firms (Weber's principle of agglomeration) and customers (Zipf's principle of least effort). So, the most distinctive feature is that each thirty-degree sector has the same number of central places, but the volume of activities at these central places varies considerably across the economic landscape. There is added realism in that smaller places, having low functional complexity, can sometimes serve large places that enjoy more functional complexity. In short, Lösch allows a more realistic spatial division of labor on the landscape. Perhaps, Lösch's most important

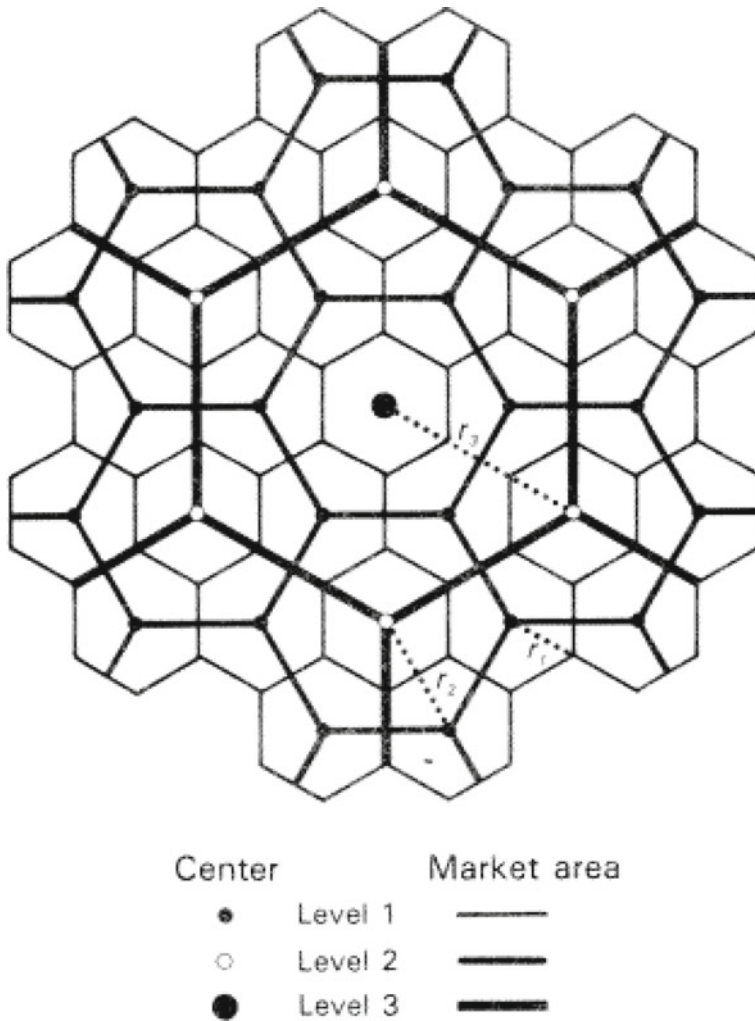


Fig. 1 A typical CPT hierarchy of hexagonal market areas (this figure is a reproduction of Figure 3.9, on page 57 of [Berry and Parr 1988](#))

contribution was his novel treatment of spatial demand where he used the calculus to aggregate household demand over space in order to compute the entire market area demand facing a firm. This allowed him to use the standard tools of microeconomics to specify the firm's profit-maximizing price and output and to specify the firm's optimizing market area. Most observers are skeptical of Lösch's grand spatial scheme, realizing he emphasized how the world should be instead of how it really is. Moreover, he was no clearer than Christaller in explaining how the atomistic behavior of firms in space can actually generate the complicated multi-good equilibrium found in so many textbooks.

3 The three key concepts

Part of the great value of CPT rests in its ability to integrate a number of key concepts in regional science, including (i) consumer choice; (ii) firm agglomeration; and (iii) the functional hierarchy. Firms offering goods and services strive to occupy locations that are central to their markets but they also cluster in order to reap mutual advantages. Dispersed consumers strive to minimize the total costs involved in purchasing these goods as firms arrange themselves in space. With various adjustments, firms eventually settle on price-location solutions that minimize the travel costs for consumers, and, at the same time, the clustering of firms leads to the formation of a hierarchy that efficiently provides all possible goods. In doing so, tier-specific bundles of goods are formed, each having a different shopping frequency. These three key concepts—choice, agglomeration, and the hierarchy—are so prevalent throughout spatial sciences that some of this review is now devoted to each.

3.1 Consumer choice

A wide literature, focusing on a few fundamental concerns, addresses the issue of how informed agents choose among alternative resource opportunities: What features of these source opportunities do the agents evaluate? What rules do agents follow when evaluating alternatives? And, what rules determine the actions of agents after their evaluations? Matters become more complicated when choices are made in a spatial setting wherein agents and resource opportunities occupy different locations. Information, which might be incomplete, tends to exhibit distance-decay as well as time-decay. Since energy or money expenditures are involved in matching agents and opportunities at the same location, behavior in space is often very complex to evaluate (see, [Golledge and Stimson 1997](#)).

Like foraging theory in ecology ([Stephens and Krebs 1986](#)), the households of CPT must decide between nearby places offering a narrow range of goods and services—and more distant places offering a larger and deeper array of goods and services ([Golledge et al. 1966](#)). Perhaps, the most novel related research is [Rushton \(1969\)](#), who constructed space-preference functions to address this fundamental trade-off. In a sense, this function extends the important notion that behavioral isolines can be identified for shopping customers ([Reilly 1929](#); [Huff 1963](#)). After collecting data on purchases, curves are drawn in two-dimensional space representing those combinations of travel distance (x -axis) and town population (y -axis) that households are indifferent to. Curves found higher and to the left (short distances, large towns) are preferred to curves found lower and to the right (long distances, small towns). Stylized curves, with no inflection points, indicate that consumers must be compensated with increasingly greater shopping opportunities for incremental travel intervals of equal length. These indifference curves suggest that distance-population trade-offs will vary between rich and poor households and, likewise, between rich and poor regions. After some inflection point (time/cost threshold), households are willing to travel long distances to take advantage of the rich shopping opportunities found at large,

diverse places. In short, customers do not shop around—a process that involves acquiring data, searching, and evaluating—for cheap, standardized, frequently demanded goods, but they do shop around for expensive, differentiated, infrequently demanded goods.

Simon (1952) was among the first to question total rationality in human actions—a notion extended by Tversky and Kahneman (1974), among others. In this spirit, many observers have outlined more realistic models of shopping behavior than those espousing strictly cost-minimizing behavior. Some of these researchers have advocated simply extending the set of attributes that households consider when making their evaluations. In addition to per unit price, other attributes of the goods and their establishments—including selection, service, safety, and congestion—are known to influence household behavior. Moreover, transportation infrastructure is hardly uniform and movement imagery also constrains shopping travel. Other research has focused on the spatial search process where repeated “shopping trials” for convenience goods are undertaken, sometimes for many weeks, before a stable pattern of shopping behavior is chosen. Moreover, there is ample evidence that households may not be aware of their full set of shopping opportunities.

In response, Louviere (1974) outlined an information-processing paradigm to take into account many of these different facets of shopping. In a sense, this is a computational model for Simon (1952) satisfying behavior—one in which all of the key attributes of goods or situations are evaluated with trade-off functions. This information is weighted to arrive at a single numerical score that informs the household’s shopping decision. However, it seems that this approach, like its various counterpart methodologies, has been used more to assess how specific attributes of shopping centers, including those of anchor tenants, affect consumer choice instead of how households substitute travel distances for changes in these other location-specific attributes (Finn and Louviere 1996).

3.2 Firm agglomeration

CPT addresses a number of decisions that are simultaneously or sequentially entertained by spatial firms. To begin with, the firm must calculate the optimal price-quantity combination under conditions of free entry and exit. At the same time, an optimizing location for the firm must be identified where, via spatial competition, the firm is maximally separated from its nearest rivals. However, firms typically cluster in space with firms in other tertiary industries, so they can enjoy the benefits of urbanization economies (Jacobs 1961). Moreover, in larger places, these same firms cluster with other firms in the same industry if local demand is sufficient to allow the multiplication of establishments, and localization economies will also emerge. Some notable empirical studies have addressed the relationships between price and location in this context (Haining 1983; West and von Hohenbalken 1984; Fik 1988). This idea of spatial agglomeration, or geographic clustering, is shared by many other sciences: for example, densely packed, interacting agents are believed to promote creativity and to spur on the innovation and diffusion of new ideas and skills (Hall 1998; Glaeser and Maré 2001; Fujita 2009).

In economics, spatial clustering usually reflects external economies of scale, which are known to arise naturally in a variety of ways (Marshall 1920; Krugman 1991a,b; Rosenthal and Strange 2006). First, the agglomeration of people and firms leads to knowledge spill overs where both patents and innovations are themselves clustered, usually in cities—so success tends to breed further success. Second, agglomeration tends to promote input sharing where firms learn to outsource to common local suppliers. Concentrated industries have a higher degree of purchased input intensity than scattered industries, meaning that behavior favors the emergence of other firms in the same district operating along the same commodity chains. Third, agglomeration promotes labor pooling, thereby allowing more efficient matches between workers and employers. Agglomeration also arises because of multipurpose and comparative shopping, primarily in the retail and service industries. Multipurpose (including multistep) shopping allows households to reduce per unit transportation costs, while comparative shopping reduces search costs when household information is incomplete (Thill 1986; Eaton and Lipsey 1979). Both types of demand externalities are recognized in the competing destinations model, which also might reflect hierarchical organization (Fotheringham 1983; Fik and Mulligan 1990). Besides these benefits, agglomeration also ensures longitudinal stability in economic communities. As aging industries die, they are replaced by new industries, though external economies of scale facilitates this transformation. Both North (1955) and Blumenfeld (1955) noted this important role of geographic clustering in their early discussions of economic base.

Although his discussion is constrained to manufacturing, Arthur (1994) suggests that there are two somewhat different perspectives on the formation of economic landscapes. One is deterministic, where firm sites are chosen because of scarce geographic endowments, local rents, or finite shipment possibilities. In this sense, today's industrial landscapes might be very different from earlier ones. The other is probabilistic and, here, path dependency in firm location is crucial—today's industrial landscapes, for example, are very much inherited: late-arriving firms are attracted to early-arriving firms even if the original location decisions were not optimal ones. As a result, contemporary industries might still remain competitive even though they do not occupy the most favorable of all possible locations.

Classical CPT follows the former school of thought, where the foundations of the settlement system are those small clusters of tertiary firms, evenly spaced across the landscape, that offer goods and services to a uniform population of rural households. Different-sized clusters of firms are then essentially stacked on this functional base one level at a time. But clusters arise only at those locations that prove to be locally optimal for (lower-level and same-level) consumer purchases—and they do not materialize by chance: in fact, any locations that are poor for business are quickly eliminated. So in smaller central places, agglomeration is largely of the simple urbanization variety where entirely different convenience functions are offered as simple bundles. But in larger central places, there is a multiplication of establishments along the same lines, so the functional agglomeration includes both the localization and urbanization varieties. Various demand externalities inherent to shopping also promote firm agglomeration at higher levels of the hierarchy, thereby concentrating industries and distorting the city-size distribution of the entire central place system.

3.3 The functional hierarchy

Another wide literature in the natural and social sciences makes use of the concept of hierarchy, where the separate levels of a complex entity are of conceptual and operational interest. This framework is often visualized as a system having hierarchical—and self-replicating (Batty 2007)—properties that serve to maintain order and promote efficiency. These properties either have emerged through natural agency or been adopted through human agency. In many cases, a hierarchy appears because of systemic self-organization.

The evidence for hierarchies is widespread. Christian (2004) notes that both top-down theories of state formation, which emphasize coercion, and bottom-up theories of state formation, which emphasize coordination, account for the importance of hierarchies. These organizational tiers seem to be indispensable in the management of positive feedback loops—mechanisms linking population growth, collective learning, and technological innovation—thereby allowing societies to transition to higher levels of complexity. Moreover, hierarchies are adopted as mechanisms to support those norms that reduce conflict between agents or social groups (Axelrod 1997). Urbanists like Jacobs (1961), Lynch (1981), Friedmann (1986), and Sassen (1991) have discussed the incidence of hierarchies at several geographic scales. At one scale, they point out that certain cities like New York, London, and Tokyo are at the top of a global city hierarchy because they dominate as business headquarters, financial industries, and media centers. But, at another scale, they recognize that a hierarchy is embedded in a wide variety of private and public agents that are situated in these global cities. At this scale, some sort of tiered organization is vital for internal control of firm activities and for transnational coordination with other firms and governments. Williamson (1975, 1985) described the role of hierarchies in economic institutions, where the degree of hierarchy is closely tied to the nature of decision-making and the formulation of contracts. Hierarchy is very great when the responsibility for making decisions or negotiating contacts is concentrated in a few agents; however, when decisions and negotiations are made by many individual agents or are subject to collective approval, then hierarchy is not so great. In this literature, the twin notions of hierarchy and inequality appear to go hand in hand.

4 The emergence of a functional hierarchy

The most serious shortcomings of CPT are its somewhat weak behavioral explanation for the emergence of the hierarchy and its lack of a solid microeconomic foundation. Even the simpler Christaller scheme has some difficulties in providing a convincing story for the emergence of the regularly spaced, clustered businesses based solely on the atomistic behavior of firms and households. Three somewhat different approaches are now discussed: one economic and the other two evolutionary.

4.1 Spatial evolution

A somewhat different perspective on the emergence of a functional hierarchy has its roots in dynamic modeling, and White (1974, 1977, 1978) was evidently the first to

explore this sort of thinking in a CPT framework. Numerous simulations for central place populations were run where coefficients varied for population multipliers, household transportation rates, and the fixed costs of starting up businesses. The simulations show that, when transportation costs are low, centrality beats other factors and high-order places flourish—but when these costs grow, the distance to neighboring centers increasingly constrains household travel, restricts high-order growth, and flattens the overall city-size distribution. At the same time, larger centers are relatively insulated from chance downturns by the high fixed costs involved in the provision of high-order goods. Under varying parameters, a spatial equilibrium resembling Christaller's logic invariably emerges; but the balance between transportation costs and indivisible production costs is key to determining the final distribution of population sizes. In these models, population imbalances in CPT systems are especially sensitive to high-order population multipliers. However, the results endorse systemic stability in that the simulated equilibria do not seem to depend upon the initial locations or sizes of central places. Positive feedback loops endogenously link together growth processes and growth constraints.

Allen and Sanglier (1978, 1979) visualize a mixture of probabilistic and deterministic forces guiding the evolution of central place systems. Entrepreneurs are thought to calculate aggregate demand for new products, and this figure is translated into potential employment by a logistic-type function. When two (bundled) goods are provided, a wide number of stable solutions emerge as centers grow or decline according to their adaptive capacity. But when a third bundle is added, the positive feedback loops related to spatial agglomeration come into play. This function might be allocated anywhere in the system, but it will only survive (and satisfy certain threshold requirements) when it is allocated to a point already housing the second bundle. The winning combination of a central location and a substantial local population proves to be critical for the survival of this third-order activity. In their later work, Allen and Sanglier (1981a,b) introduce commuting or crowding into this spatial model of systemic evolution. In this framework, growth and change throughout the spatial system is seen to pass through four distinct stages: (i) initial allocations of population and employment but only to certain points in space; (ii) subsequent population decentralization due to crowding; (iii) induced economic decentralization out from core areas; and, finally, (iv) ongoing competition between a few major nodes of activity. This line of research was promising because it dealt with both the intra- and intercity forces that shape the evolution of urban and regional systems.

4.2 Microeconomics

While CPT endorses the agglomeration of firms, it also postulates that these different-sized clusters of firms tend to be distanced from one another. This dichotomous behavior has its roots in Hotelling (1929) and Smithies (1941), who recognized that, even in the face of constant customer densities, firms can—and do—switch their incentives for location, depending upon extant market conditions. Following Lösch's analysis of spatial demand in a single industry, a pool of research aimed at examining the competitive behavior of spatial firms (Norman 1986) developed. Here, it was soon

realized that the market, or aggregate, demand facing each firm was comprised of three adjacent forms: (i) monopoly, (ii) competitive, and (iii) super-competitive (Beckmann and Thisse 1986). The price-elastic household demand would always provide sufficient incentive for firms to seek middle locations between their spatial rivals, thereby protecting their spatial monopolies.

From this point of departure, two different perspectives on spatial competition arose. From the first perspective, price behavior was really the sole consideration, where identical firms were evenly dispersed through unbounded spatial markets. In fact, the same mechanism outlined by Lösch was used to drive firms toward a stable equilibrium—presumably they would continue to enter the market until they were packed so closely together that no firm could earn an economic profit; at that point, the entire market would be in a stable equilibrium (Beckmann 1971; Stern 1972). From the second perspective, firms were actually embedded in space where they would make explicit price and location decisions with respect to nearest rivals (Eaton 1972). In this context, the entire market could be either unbounded or bounded in space, a feature that added both complexity and realism to the analysis (Mulligan and Fik 1994, 1995). A stable equilibrium was only reached when no firm had any incentive to change its price-output-location decision. Firms could earn a pure profit—but only as long as this amount was insufficient to entice new entrants. However, under fairly reasonable entry scenarios, a stable equilibrium will not always materialize in such markets (Eaton 1976; Eaton and Lipsey 1976). In fact, various perturbations can destroy the regular spacing of firms and allow those firms to dominate market areas of different sizes. Spatial asymmetry can also occur if fixed costs are separated into sunk costs and contractual obligations, where the former can restrict the free location adjustment of firms.

However, there have been only a few attempts made to construct a multi-good equilibrium based on behavioral rules. One approach, by Eaton and Lipsey (1976), tries to clarify the microeconomic underpinnings for central places by highlighting the demand externalities of consumer shopping. By recognizing the inherent indivisibilities of goods, adopting constant transportation rates, and then introducing multipurpose shopping as a motivation for firms to cluster, they were actually able to develop various types of two-level hierarchies in one-dimensional markets. Here, Christaller's system was one of the multiple spatial equilibria that could arise. It is argued that, as long as single- and multipurpose shopping coexist, higher order places are always able to steal potential demand from adjacent, lower-order places. This means that, as long as the location of firm clusters is endogenously determined by competitive forces and not chosen by a planning agency, spatial asymmetry will often characterize the market equilibrium.

Yet another approach—the two-good cost-minimization model developed by McLafferty and Ghosh (1984, 1987)—remains an admirable attempt to address the agglomeration versus separation issue in two spatial dimensions. Using a location-allocation model designed for public facilities, their model examines all possible configurations of firm location under conditions of multipurpose shopping for dispersed households. Each household's decision-making calculus is embedded in the various firm locations, where a mixture of single- and multipurpose shopping trips is chosen. As in Eaton and Lipsey, clustered low-order firms can always steal revenues from

their non-clustered low-order rivals and some of these non-clustered firms might even disappear from the landscape. A large batch of numerical simulations indicates that multipurpose shopping invariably promotes the symmetrical hierarchy of city systems in a manner consistent with Christaller's logic. Depending on various price and cost circumstances, households found at different locations might change their mix of single- and multiple-purpose shopping trips—but the firms that service those households always appear to agglomerate only at certain key locations on the landscape. The authors also point out that such agglomeration is strengthened by opportunities for comparative shopping.

4.3 Dynamic extensions and the new economic geography

CPT and the NEG have many similarities, though they differ in some key ways.¹ Given space constraints, the discussion is constrained to only briefly describe these features. Readers interested in more detailed comparisons are referred to the numerous sources cited here—in particular, to the [Brakman et al. \(2009\)](#) NEG textbook—and survey of agglomeration economies in general can be found in [Puga \(2010\)](#).

The key advantage of CPT is it can explain existing urban hierarchies including the geographic spacing of cities and their hinterlands, plus why various goods and services are found in particular locations. The primary criticism of the framework is that its models typically lack microeconomic foundations, are primarily geometric, are partial equilibrium ([Fujita et al. 1999](#); [Tabuchi and Thisse 2011](#), p. 241; [Brakman et al. 2009](#), pp. 44–46), and efforts to make them dynamic are often *ad hoc* ([Partridge 2010](#)). Some of these criticisms are unfair because, as just described in earlier sections, microeconomic foundations, in particular, have received considerable attention for many decades.

For these and other reasons, NEG formulations became popular after [Krugman \(1991b\)](#) seminal paper and CPT research faded from the limelight. The base NEG model following [Krugman \(1991b\)](#) typically has two regions with a traded sector (manufacturing) produced in a Dixit–Stiglitz monopolistic competitive market with multiple varieties, increasing returns to scale, and labor is the only input. There is a non-traded sector (farming) produced in a perfectly competitive market with constant returns to scale. Utility is a nested Cobb–Douglas function with manufacturing demand for individual varieties being determined by a Dixit–Stiglitz CES utility function. There are no transportation costs for food, while manufacturing transportation costs are an iceberg function. Manufacturing labor is mobile, in which labor moves to the region offering the highest real wage. Later NEG models kept much of the same form but added other features that somewhat improved the realism of the model such as adding a non-traded housing sector (see [Brakman et al. 2009](#) for more details). The key advantage of the NEG model is that it is an endogenous model in which dynamics can be traced through in simulation form.

¹ As pointed out by an anonymous referee, one distinction between CPT and NEG is that CPT is a view of the spatial hierarchy, while NEG is methodologically defined.

In simple terms, the key difference between CPT and NEG is that CPT models are explicitly spatial (they can easily be shown on a map, for instance), and NEG models are typically a spatial in the sense there are usually just two regions with most of the focus placed on dynamics after changes in key parameters. Likewise, CPT focuses on the location of specific types of industries, as well as the relative returns to scale across different industries that lead to spatial concentration—that is, higher- versus lower-ordered services (though [Tabuchi and Thisse 2011](#) are a counterexample). NEG frameworks stress transportation costs and a general equilibrium formulation that produces endogenous dynamics.

NEG models address many of the shortcomings of CPT by providing microeconomic foundations that incorporate, among other things, increasing returns to scale, monopolistic competition, and a “love of variety” among firms and households (see [Brakman et al. 2009](#) for a comprehensive survey). A self-reinforcing pattern of increasing agglomeration can dynamically take hold due to the advantages that suppliers and customers have as a result of close access to inputs and outputs, and larger cities have more variety that attracts firms and households. In most NEG models, falling transportation costs induce more concentration so that industries can take advantage of the increasing returns to scale. Still, NEG models have shortcomings of their own ([Neary 2001](#); [Nijkamp 2000](#); [Ottaviano et al. 2002](#)). Some of these include restrictive production functions and the tendency for knife-edge, or “catastrophic” results that are sensitive to small parameter changes. For example, the slightest changes in transportation costs in the base models can sometimes lead to everything from full agglomeration in either region or equal shares in either region. The increasing returns assumptions suggest that large shocks may lead to catastrophic consequences that are not generally supported by empirical work based on natural and man-made disasters.²

Another criticism of NEG models is the use of unrealistic assumptions such as iceberg transportation costs ([Neary 2001](#); [Nijkamp 2000](#)), which is done for analytical simplicity. For example, [Bosker and Garretsen \(2010\)](#) show that even small changes in how transportation costs are specified in empirical NEG models can lead to wildly different implications. There are also important theoretical shortcomings from iceberg transportation costs. [McCann \(2005, 2007\)](#) and [Fingleton and McCann \(2007\)](#) show that iceberg transportation costs are inconsistent with the distance cost pricing for logistics and transportation, as well as for explaining the movement people and ideas, which theoretically requires square root terms to be consistent with transaction costs models (also see the discussion in [Bosker and Garretsen 2010](#), pp. 499–501).

Perhaps even more problematic about NEG models is that, when realistic factors such commuting or amenities are added into NEG models, their complexity quickly becomes overwhelming ([Tabuchi and Thisse 2002, 2006](#)). For example, [Partridge \(2010\)](#) argues that NEG models do a poor job of describing contemporary growth processes in the United States, compared to other explanations such as amenities.

² For example, [Davis and Weinstein \(2002\)](#) find that despite the heavy bombing that Japanese cities withstood during World War II, they had significantly recovered by the end of the 1950s. [Bosker et al. \(2007\)](#) found a similar pattern when considering the devastating effects of Allied bombing and military campaigns on the German urban system. [Xiao \(2011\)](#) is an example of a study that finds that natural disasters such as floods have generally short-term effects on regional economies.

Krugman (2010) notes that NEG models are better at describing developing country dynamics such as in China or India (see Hering and Poncet 2010 for an application to China).

Trying to overcome some of criticisms of CPT, Fujita et al. (1999) adopt an NEG model to show how a CPT hierarchy can arise during initial territorial settlement—though one weakness of their model is that it does not lead to the spatial layering of services that defines CPT. Tabuchi and Thisse (2011) add further richness to the Fujita et al. (1999) framework by showing how an urban hierarchy can develop in which larger central places have more industries and higher-ordered services, and where the very top of the hierarchy is persistent. Yet, their representation does not capture empirical regularities such as Zipf's law, which would give it more realism. Recent work suggests that CPT has some significant advantages for describing such empirical regularities compared to other approaches. For example, Hsu (forthcoming) uses an equilibrium entry model—with micro foundations—to derive a CPT representation that (i) meets the CPT hierarchy property (higher-ordered cities have the same industries as in lower-level cities and more); (ii) is consistent with Zipf's law, and (iii) is consistent with the recently discovered *number-average-size rule* (Mori et al. 2008) that relates the number of cities in which an industry is located with the average population of those cities. Further, Hsu and Holmes (2009) use a dynamic programming model to show that Hsu's (forthcoming) model can lead to optimal results from a social planners' perspective. While it may be too early to claim a rebirth in CPT as the leading model of urban systems, the inflexibility and lack of spatial realism in NEG and other frameworks are spurring renewed interest in the older framework.

5 Growth and development

The predictions resulting from CPT have strong implications for the growth and development of a nation's urban system and the development of its regions. This section briefly describes empirical tests of CPT for across several dimensions—the spatial structure of industries, the stability of the urban hierarchy, growth across the hierarchy, rural–urban interdependence, and commuting. Generally, the first two subsections stress the development of larger central places and subsequently smaller cities, while the later two-subsections relate to intraregion dynamics and regional growth in the CPT framework. One caveat is the general literature is immense, and so the discussion is constrained to the more recent North American literature.

5.1 Industrial structure across cities

The key distinction between CPT and other urban system theories is that CPT places the spatial nature of the system at the forefront. Thus, while other urban system theories also describe city types, size distribution, and industry structure (Henderson 1974; Gabaix 1999; Duranton and Puga 2001; Eeckhout 2004), the discussion here focuses on those studies with a strong spatial framework drawn from CPT.³

³ Note that Beckmann (1958) did tie the power law for city-size distributions to CPT, but the framework lacked an economic model (Hsu, forthcoming).

While standard CPT is static, previous sections have described how extensions of it imply dynamic growth patterns. Because the highest tiered central places attract industries with the greatest scale economies, proximity to the highest tiered central places allow nearby locations—including smaller communities—to take advantage of close access to these services, allowing them to grow faster. Olfert and Stabler (1999) note that input–output linkages also greatly favor the top of the hierarchy. Using Canadian data from Saskatchewan, they find that an exogenous increase in expenditures in the lowest tier place leads to almost equal amounts of induced expenditures in the highest tiered city in that region—a sign of the high leakages for the lowest to highest tiers. Overall, they report that multipliers are about double in the highest tiered central place as in the lowest tiered place. The advantages found in highest tiered cities underlie Polèse and Shearmur (2004b, 2006) examination of the relative concentration of industry types across the urban hierarchy.⁴ In defining a core/periphery delineation, they divide the Canadian urban hierarchy into tiers based on metropolitan area population (including a rural tier) and distance to a highest tiered metropolitan area (more than 500,000 population). Over various periods between 1971 and 2001, they consistently find a cross-sectional pattern wherein high-order services concentrate in core cities and nearby locations. Likewise, this pattern is very stable and may be accelerating over time.⁵ However, there is industry churning: as advanced industries become standardized, they are crowded out of the core central places due to cost considerations and commonly end up relocating to more remote locations.

Using data from the United States, Klaus and Fafchamps (2006) find that service industries are generally concentrating—but, also, that, as manufacturing becomes increasingly standardized, it is deconcentrating. However, Partridge et al. (2008a) find that remoteness has neutral sectoral effects—or, in other words, that industries are concentrating about equally.

Probably the most interesting recent work for testing the industry location features of CPT has revolved around the *number-average-size rule* (NAS) (Mori et al. 2008). As noted above, NAS relates the number of cities in which an industry is located with the average population of those cities. As shown first by Mori et al. (2009) and then by Mori and Smith (2011) with a more advanced clustering approach, the NAS rule holds for Japan, whereas Hsu (forthcoming) shows that it also applies for the United States. Moreover, the NAS rule is closely associated with Zipf's Law (Rank Size Rule) and the hierarchy principle that forms the core of CPT's tiering of cities in which subsequently larger cities have higher-ordered services (Mori et al. 2008; Mori and Smith 2011). This recent CPT empirical work illustrates that the industry hierarchy effects that underlie CPT is at work at describing central places up and down the urban hierarchy. More work is needed to assess how the spatial disposition of industries is evolving in advanced economies, but CPT remains an excellent guide.

⁴ Polèse and Shearmur's industry breakdown tends to be on the basis of demand for specialized labor and market threshold effects (e.g., Wensley and Stabler 1998; Shonkwiler and Harris 1993).

⁵ Polèse et al. (2007) find a similar pattern for Spain.

5.2 Stability of the urban hierarchy

The CPT agglomeration patterns described above suggest that upper-tier central places have advantages that allow them to persistently remain near the top of the hierarchy, while there is likely more churning near the bottom because lower-tiered places possess fewer agglomeration advantages. Indeed, [Black and Henderson \(2003\)](#) conclude this pattern applies for the United States. In examining decade-by-decade data over the 1900–1990 period, they find that the urban hierarchy is relatively stable—though with some churning—but the top is especially stable ([Dobkins and Ioannides 2001](#)). [Polèse and Denis-Jacob \(2010\)](#) examine the stability of the top-ten cities in the urban hierarchy across 74 countries over the twentieth century. They find that the placement of these cities in the hierarchy is very stable, illustrating the inherent advantages that large cities have in maintaining their dominance. Their finding suggests that American patterns such as the rise of Sunbelt cities (for example, Atlanta, Dallas, and Phoenix) in the late twentieth century are more of the exception on an international basis. Likewise, reversals of primate cities (for example, Toronto and Montreal) are rare and are functions of major political or cultural shocks, not economic forces ([Polèse and Shear-mur 2004a](#)). Yet, a key implication of findings that the hierarchy is stable is that policy efforts to help lagging regions will likely fail unless the underlying dynamics that produced the urban hierarchy are altered. It is worth pointing out here that some analysts ([Ioannides and Overman 2004](#), p. 132) have commented that “the precise implications” of NEG models “for growth in a system of cities are pretty fuzzy.” Though some progress has been made—see [Fujita and Mori \(1997, 2005\)](#), [Mori \(1997\)](#), [Fujita et al. \(1999\)](#), [Overman and Ioannides \(2001\)](#)—it seems to the present authors that CPT, with its built in systems dynamics ([Berry 1964](#)), may, in fact, provide a more straightforward conceptual foundation.

5.3 Regional growth in a CPT framework

The above two sections discussed empirical studies about the location of cities and the stability of the urban system in a CPT framework (especially at the top), but it did not consider how CPT can explain growth dynamics at a regional and national scale. Such research was hampered in early CPT research because of the primitive nature of geographic information systems (GIS) technology. The lack of GIS technology limited empirical efforts to examine the effect of CPT across an entire economic landscape, like the United States or Europe. Advanced GIS technology has changed this. For example, [Partridge et al. \(2008a,b, 2009\)](#) use a CPT-based framework to assess county-level economic activity across the entire continental United States. Their urban hierarchy from top to bottom also includes rural areas and is similar to that of [Polèse and Denis-Jacob \(2010\)](#). Partridge et al.’s innovation is that they explicitly introduce distance costs to reach all of the higher-tiered urban areas for each county, particularly accounting for the *incremental* distance costs for reaching each succeeding higher tier to obtain services that are not available at the subsequent lower tier, which is consistent with a CPT framework. [Partridge et al. \(2008a,b\)](#) find that consistent with agglomeration predictions from CPT, employment and population are moving closer to higher-tiered

urban areas, though not necessarily within them. Moreover, [Partridge et al. \(2009\)](#) find that housing prices and wages are following a similar pattern. [Partridge et al. \(2008a,b\)](#) find that these agglomeration forces accelerated after 1950 and so argue that twentieth century improvements in transportation and information technology favored the core, not the periphery. [Wu and Gopinath \(2008\)](#) also draw similar conclusions using a definition of remoteness based on adjacency to metropolitan areas in an empirical framework that uses spatial econometrics.

Because market potential plays a much smaller explanatory role than incremental distance in Partridge et al.'s results ([2008a; 2008b](#)) empirical results, they downplay other explanations such as NEG because the latter would predict that market potential is the driving factor, while CPT predicts that it is the geographic proximity in the hierarchy that matters. Likewise, because other agglomeration explanations—including knowledge spill overs—are more geographically limited (e.g., [Rosenthal and Strange 2008](#)), this suggests that explanations within a CPT framework are probable causes. Similarly, [Portnov and Schwartz \(2009\)](#) show that European growth is also consistent with CPT representations, though their analysis does not explicitly rule out other possible explanations.

5.4 CPT and economic development policy

The CPT framework has long been seen as an imperative for policymakers and other actors in functional economic regions to cooperate in public service delivery, infrastructure provision, and economic development. This call makes economic sense when there is a need to internalize spillovers and when there are network effects and economies of scale. In particular, the growing importance of agglomeration forces—as implied in [Partridge et al. \(2008a,b\)](#) and [Wu and Gopinath \(2008\)](#)—along with rising industry threshold effects—suggest that rural areas have much to gain from working on regional basis with urban areas ([Fox and Kumar 1965; Berry 1970; Wensley and Stabler 1998; Olfert and Stabler 1999; Partridge and Olfert 2011; Carruthers et al. forthcoming; Castle et al. 2011](#)).

Modern notions of regional economic development using CPT differ from how the framework was originally envisioned by [Christaller \(1933\)](#) and [Lösch \(1941\)](#). Specifically, rather than input–output linkages between and among firms and households, contemporary regional linkages are arguably more commonly established via commuting patterns. In particular, this trend has accelerated through better highways and employment declines of rural primary sector industries due to labor saving productivity growth. Thus, it is often the case that the best economic development strategy for lower-ordered locations is to increase their quality of life to attract migrants from urban centers and commuters that are tied to them ([Henry et al. 1997; Renkow and Hoover 2000; Carruthers and Vias 2005](#)). Likewise, growth center approaches that stress growth in the highest level tier are based on the premise that the best rural development strategy is an urban-led strategy that takes advantage of the commuting linkages ([Ali et al. 2011; Goetz et al. 2010](#)). [Fox and Kumar \(1965\)](#) and [Ali et al. \(2011\)](#) are examples that illustrate the utility of CPT for understanding rural–urban regions based on commuting. While there are strong

arguments for collaboration within such regions, there is a need for further research as to whether these regions should form an overarching regional government or whether it is best to leave multiple actors in place even if it leads to fragmented governance—that is, what is the proper centralization in regional governments, if any (Haughwout 1999, 1997)? Examples of analyses exploring the spatial consequences of different arrangements of land use governance include Carruthers (2003) and Ulfarsson and Carruthers (2006).

6 Planning

The United States has no national urban land use policy, which makes it all the more imperative that the market-driven evolution of its urban and regional systems is understood on its own terms: planners and other policymakers require knowledge of both inter- and intraregional dynamics in order to organize their thinking and take appropriate action. This section sets out some of the ways in which CPT enters the planning arena and informs planners' decision-making processes both across and within regions. Along the way, it furthers a central argument of this paper—that, at least from a normative standpoint, NEG and other “competing” models should be viewed as complementary, not alternative, frameworks.

6.1 Interregional planning

As just noted, in the United States, there is no territorial urban policy—land use is generally governed through so-called home rule authority, meaning at the most local of levels—so, at present, there is no direct role for interregional planning, *per se*. That said, planners do regularly engage in urban and regional economic development, plus in the process of ensuring that functional economic regions, like those discussed throughout this paper, perform effectively and interact effectively. Moreover, activist organizations, like the Regional Plan Association⁶ (RPA) of the New York–New Jersey–Connecticut metropolitan area, advocate for adopting an approach to planning that is essentially organized around the spatial hierarchy of CPT. For example, America 2050,⁷ a project of the RPA, has identified ten emerging mega-regions that it argues for treating—from a policy perspective—as functional economic regions: (i) the Northeast, (ii) the Great Lakes; (iii) the Piedmont Atlantic; (iv) Florida; (v) the Gulf Coast; (vi) the Texas Triangle; (vii) the Arizona Sun Corridor; (viii) Southern California; (ix) Northern California; and, finally, located in the Pacific Northwest, (x) Cascadia. According to Pierce and Johnson (2008, p. 245), these regions are directly analogous to “the Global Integration Zones identified by European and Asian planners [and] all show strong linkages through overlapping suburbs, increased commuting between metro areas, strong transportation connections, and similarities in their industrial clusters.” Figure 2 presents a map of the American constellation of

⁶ See: <http://www.rpa.org>.

⁷ See: <http://www.america2050.org>.



Fig. 2 The American constellation of core-based statistical areas

core-based statistical areas following [Pierce and Johnson \(2008\)](#). Although they do not invoke CPT by name, these policy entrepreneurs are referring directly to the world it describes when they argue for an expanded role for regional planning on the basis of functional economic regions that are embedded within a national hierarchy.

And, this kind of thinking is more than just theoretical: at the time of this writing (fall 2011), three cabinet-level federal agencies in the United States—the Department of Housing and Urban Development, the Department of Transportation, and the Environmental Protection Agencies—are engaged in the Partnership for Sustainable Communities,⁸ an ambitious effort that explicitly recognizes the need for interregional planning. Among other activities, in 2010 alone, this program distributed more than \$400 billion in grants, many of which were dedicated to regional planning activities. In this very tangible sense, CPT represents a theoretical framework capable of guiding planners and other policymakers as they take on problems that are increasingly complicated and interconnected across political jurisdictions—and often involve multiple metropolitan areas.

Although it is true that NEG models are capable of guiding some forms of public policy (see, for example, [Baldwin et al. 2003](#)), these models tend to be aspatial (as already discussed) and, more even problematic, analytically difficult to the point where they lack practical relevance to policymakers engaged in day-to-day planning. Among the key strengths of CPT are its intuitive appeal and ability to describe why—plus how, where, and when—systems of cities materialize on the economic landscape to both technical and general audiences. It certainly cannot do everything that many social scientists demand of a holistic economic theory, but there is no disputing its utility for providing general explanations for, and predictions of, otherwise intractably complex real-world phenomena. For this reason, it seems that, instead of competitors,

⁸ <http://www.sustainablecommunities.gov/pdf/FactSheet.pdf>.

CPT and NEG models may, in fact, be complements capable of contributing different forms of insight useful for different purposes.

This point is an important one because, so far, NEG models—for all of their power and elegance—have had little to offer in terms of practical advice for urban planners and policymakers working on interregional issues. It can be hard to derive testable hypotheses from NEG models (Head and Mayer 2004), and, for this reason, there are few, if any, direct lessons for planning in any of its recent advances on urban systems—see, for example, Fujita and Mori (1997, 2005), Mori (1997), Fujita et al. (1999). And, for this reason, it is unfortunate that, as a matter of scientific research, NEG models all but completely eclipsed the CPT framework for a 20-year period. (According to Google Scholar, Krugman's monograph, *Geography and Trade*, has been cited 7,097 times since it was published in 1991, compared to the 2,025 that the English version of Christaller's monograph, *Central Places in Southern Germany*, has been cited since being published in 1966.)

As Smolin (1997) forcefully argues in *The Trouble With Physics* (referring to super string theory), trends in scientific research can become as destructive as they are productive when certain theories are pursued to the point where they end up closing out other useful theories.⁹ None of this is to say that CPT is the perfect tool for the advancement of an interregional planning agenda—just that, from an applied perspective, the regional science community should reconsider what the framework has done to advance the pool of knowledge on development patterns and processes across regions before dropping it on account of its ad hoc nature and other shortcomings. In sum, no theory is perfect and it seems best not to discard (or disregard) CPT even if, as NEG proponents (Fujita et al. 1999, p. 27) suggest, it is “at best a description, rather than an explanation, of the economy's spatial structure.”

6.2 Intraregional planning

As stated at the onset of this section, the United States has no territorial urban policy, so planning is an expressly local enterprise that takes place within regions—and, in the main, individual communities. Urbanization is inherently complex, but it is nonetheless systematic, consisting of layers of patterns that are repeated across multiple, tiered scales of the sort that CPT describes (see Batty 2007). Thus, in developing a normative theory of urban form, Lynch (1981) emphasizes physical regularities that emerge across the spatial hierarchy of development—or, as Berry (1961) famously pointed out: just as there are systems of cities, the cities embedded in these systems are systems themselves. This stylized fact—clearly illustrated in Fig. 3, a map of the contemporary Northeast megalopolis (Gottmann 1961)—holds many practical implications for planners, some of which are outlined below.

Foremost, the most obvious application of CPT in the practice of intraregional planning is that it describes the general frame of urban spatial structure (Anas et al.

⁹ In the natural sciences, general relativity and quantum mechanics, inarguably two of humanities crowning achievements, famously contradict one another—meaning that one or both is somehow incorrect on a foundational level, but, even as scientists drive toward a “unified” theory, few seriously suggest throwing either one out.

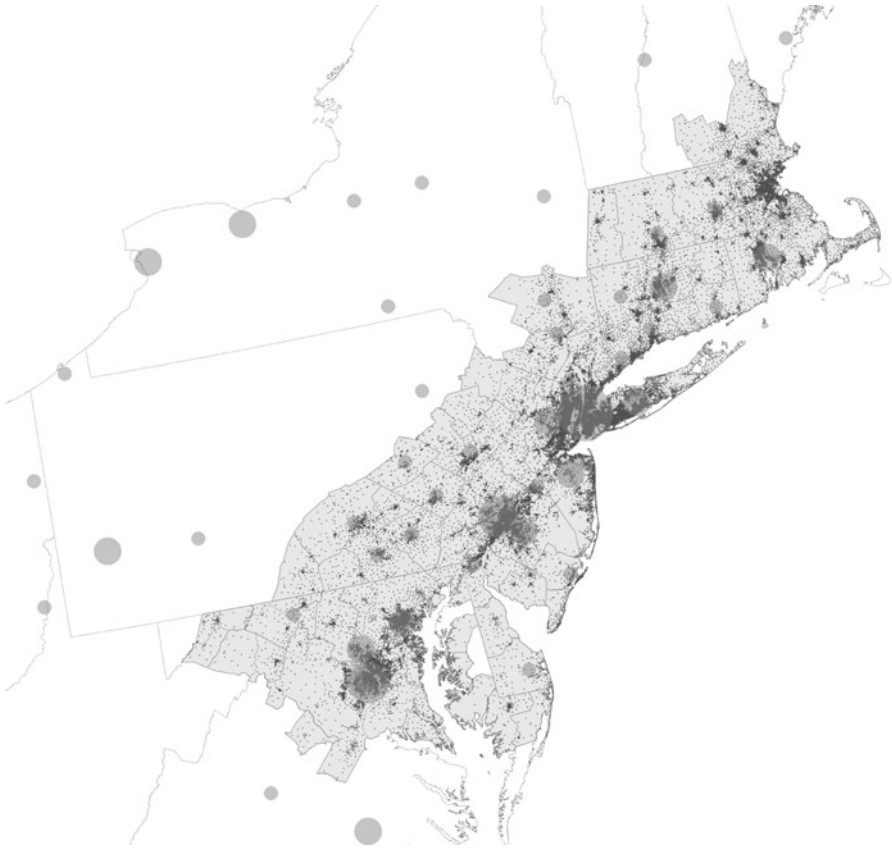


Fig. 3 The Northeast megalopolis

1998) both through time and across geographic space. For more than 100 years, dating to at least [Howard's \(1902\) *Garden Cities of Tomorrow*](#), planners have thought about urban form as a functional hierarchy and have set out transportation networks and pursued policies aimed at shaping settlement patterns accordingly. In this sense, the fundamental critique mentioned above—that CPT is a description, not an explanation, of the economic landscape—holds less relevance to planners, who need to understand the nuts and bolts of how cities are put together, than it does for theorists, who ask more fundamental questions about why there are cities in the first place. For example, in their classic paper, [Friedmann and Miller \(1965, p. 314\)](#) suggest that the city should not be viewed as a monolithic “physical entity,” but, instead, as “a pattern of point locations and connecting flows of people, information, money, and commodities.” This claim was prescient in many ways and has recently been used as a point of departure for examining recent land use change throughout the United States ([Carruthers et al. forthcoming](#)). So, on a basic level, CPT offers planners a powerful way of understanding the fabric of urbanization and making decisions—including transportation investments, allocating land use, and more—affecting the overall functionality of the regions in which they operate.

Moreover, CPT's traditions of market and locational analysis hold important implications for the siting of centralized facilities, ranging from shopping centers to water treatment plants. For example, the last major text on CPT (Berry and Parr 1988) was devoted entirely to describing the nature of market areas and what kind of services should be located where—and at what (spatial) frequency. Knowing how to evaluate such issues is of no small consequence to planners—or, for that matter, real estate developers—whose decisions on zoning and other factors that affect retail and business location impact not only the level of access to goods and services that residents enjoy, but numerous other matters of consequence, including access to jobs and even the strength of the local tax base (see, for example Kaiser et al. 1995; Hopkins 2001).

At a regional level, some of the first growth management, or smart growth, initiatives targeted “developments of regional impact,” or, in other words, large, centralized facilities whose replication would result in bifurcated demand and other inefficiencies (Popper 1981). Such interventions are highly controversial (Fischel 1989); however, so, in order to be credible, planners require a basis for understanding the nature of the organizational hierarchy of the economic landscapes they regulate—and, as argued throughout this paper, CPT offers an excellent means of doing so.

Further, a similar logic extends to the planning of infrastructure and public goods and services. Urban sprawl, a dominant mode of land use in the United States—and, increasingly, elsewhere in the world (Bruegmann 2005)—is commonly criticized by members of the planning profession and citizen activists for being wasteful and inefficient. Yet, it is also true that the nation is rich in territory and there is no imperative in terms of the amount of land available to promote more compact development (see, for example, Gordon and Richardson 1997). What's needed, then, are means of objectively discriminating among alternative land use outcomes (Talen and Ellis 2002), and the CPT framework provides a basis for doing so. There is evidence that low-density, spatially expansive patterns of land use are indeed more expensive to provide certain services to than high-density, compact alternatives (Carruthers and Ulfarsson 2003, 2008; Hortas-Rico and Solé-Ollé 2010) and CTP helps explain why: large facilities, when centrally sited, create efficiencies both in terms of the specific functions they fulfill—for example, water treatment (Hopkins et al. 2004)—and the infrastructure used to distribute their output (Mulligan 1991, 2000). Going back to Lynch's (1981) normative theory of urban form to criteria in particular stand out: (i) *efficiency*, which suggests cost-effectiveness is better; and (ii) *equity*, which suggests that residents living in low-density, outlying locations should not be subsidized by those living in more cost-effective high-density, centralized locations. Indeed, a violation of these principles suggests that a market failure has occurred (Brueckner 2000) and that the use of development impact fees or some other regulatory intervention may be justified.

7 Summary and conclusion

CPT has clear advantages for describing both inter and intraregional spatial patterns of economic activity and urbanization—and for informing planning and other forms of policymaking. Its main shortcomings are its supposed lack of microeconomic foundations and lack of a general equilibrium approach. Of course, one reason for this

shortcoming is that CPT is very rich in describing both industry location and settlement patterns in a manner that is explicitly spatial—its very realism is what makes it difficult to analytically model in a general equilibrium fashion. With renewed interest in CPT, there is a distinct need for further testing its empirical predictions with GIS and micro data on firm and household location decisions. Along with assessing how well CPT fits empirical regularities, there is also a need to evaluate whether or not CPT can inform forecasts of the continued evolution of urban and regional systems. In terms of policymaking, there is a need to better understand whether the functional economic areas that fall from CPT can be used for governance—and to better articulate its utility for evaluating urban form. In these ways and more the CPT framework is ripe for additional research that places it on more equal footing with its complements, such as NEG models.

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